



OPERATOR GUIDE • QUIKBOLUS

3D Printing a Patient Bolus on the Bambu Lab P1S

A practical, step-by-step settings + supports guide for the operator who is printing a bolus from QuikBolus output for the first time. Tuned for TPU 95A on an enclosed P1S with the textured PEI plate.

Issued 2026-05-23 • Applies to Bambu Studio 1.9+ on Bambu Lab P1S

1. Pick the right material

Direct-print bolus material has to do three things at once: stay soft enough to conform to the patient, hold dimensional accuracy through dozens of fractions, and sit close enough to water-equivalent for the planner's dose calculation to stand up. **TPU 95A** is the right default for almost every patient bolus.

MATERIAL	USE IT FOR	WHY / CAVEATS
TPU 95A (<i>Bambu, Polymaker PolyFlex, NinjaFlex Cheetah</i>)	Default direct-print bolus	Skin-flexible, conforms well, density ~1.21 g/cc — close enough to water-equivalent for most build-up applications. Hygroscopic; dry before use.
TPU 85A	Softer / more conformal bolus, very irregular anatomy	Significantly harder to print — oozes, jams, prints slower. Don't start here. Earn it on simple geometries first.
PETG or PLA (<i>rigid</i>)	Mold for casting Flexibolus or similar pourable silicone	Not for direct patient contact. Smooth interior surface = clean cast. Print the mold rigid; cast the patient-facing bolus from water-equivalent silicone.
ABS, PLA (<i>direct</i>)	—	Avoid. ABS warps and smells in an enclosed printer; PLA is too rigid for skin contact and not water-equivalent.

AMS warning. Bambu Lab themselves recommend running TPU from the **external spool holder**, not through the AMS. The AMS makes sharp bends that grip flexibles too hard and cause friction-jamming.

Disable AMS for the print and feed directly into the rear PTFE port.

2. Bambu Studio print profile — TPU 95A

Start from the built-in profile "0.20 mm Standard @BBL PIS 0.4 nozzle (TPU for AMS)" and override the values below. Save as a clinic-specific profile named *QuikBolus — TPU 95A — PIS* so the next operator gets the same settings.

SETTING	VALUE	WHY
Layer height	0.20 mm	0.16 for finer surface, 0.28 if speed matters. 0.20 is the comfort zone.
Wall loops	4	TPU at 2 walls feels noodle-like. 4 gives structural integrity to the bolus rim.
Top / bottom shells	5	Watertight patient-contact surface.
Sparse infill	15–20%, Gyroid	Gyroid keeps flexibility isotropic. Rectilinear is too stiff and prints harder lines you can feel through the skin surface.
Outer wall speed	25 mm/s	TPU loves slow outer walls — surface quality. Don't push past 30.
Inner wall speed	40 mm/s	
Infill speed	50 mm/s	
First-layer speed	20 mm/s	TPU adhesion is fussy. Slow first layer = no curling.
Nozzle temperature	235–240 °C	Start 235; bump 240 if extrusion stutters or inter-layer adhesion looks poor.
Bed temperature	55 °C	Textured PEI plate. Bump to 60 °C if first layer lifts at corners.
Retraction length	0.5–0.8 mm	TPU + long retraction = clog. Less is more.
Retraction speed	30 mm/s	Slow — don't snap the noodle back.
Bridge speed	15 mm/s	TPU bridges OK if slow.
Fan speed (after layer 3)	80–100 %	Helps definition. Layers 1–3 stay 0 % for adhesion.
Brim	5 mm, outer-only	TPU sometimes lifts at corners over a long print. Brim peels off cleanly post-print.
Z-hop	OFF	Common source of TPU jams. Turn it off.

SETTING	VALUE	WHY
Pressure advance	0.045	PIS default for TPU. Tune ± 0.005 if you see corner bulge (too high) or thin walls (too low).

Run flow calibration once. Bambu Studio → Calibrate → Flow Dynamics + Flow Rate. 15 minutes, done once per filament brand/spool. Saves you from under-extruded thin walls on every subsequent print.

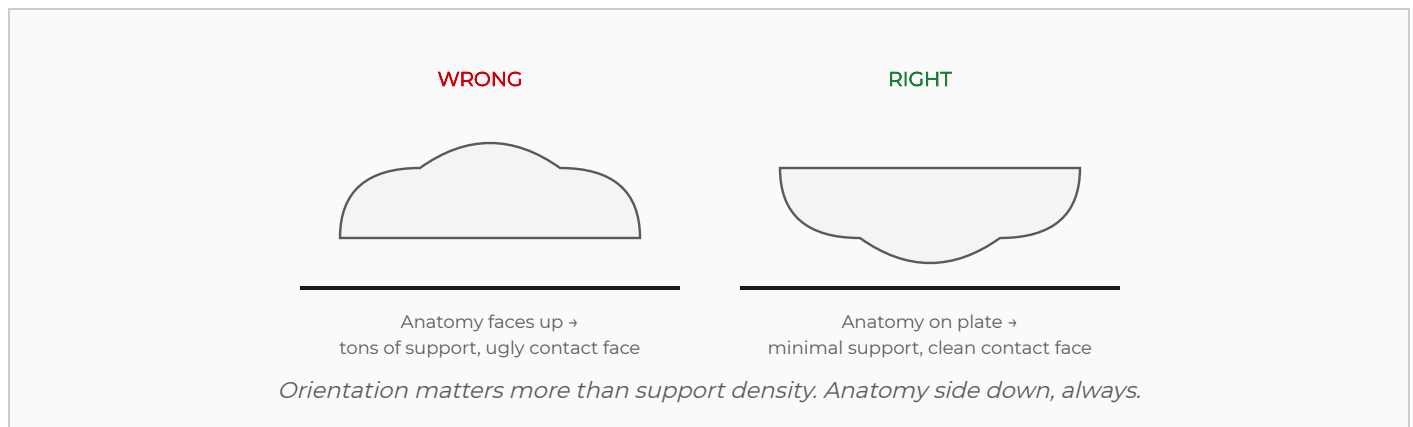
3. Supports — the part everybody gets wrong

A patient bolus is anatomical, so the surface that touches the patient is rarely flat. Some overhangs are unavoidable. The trick is not "should I use supports" — the trick is **orient the part first**, then let the supports be minimal.

Step 1: Orient the part BEFORE generating supports

This is the single most important step and most people skip it. **The patient-contact surface goes down, on the build plate.** That surface needs to be smooth and dimensionally accurate — the build plate gives you the cleanest finish you'll ever get. The outside-of-bolus surface (the one facing the LINAC) can have support scars; nobody cares.

In Bambu Studio: don't trust *Object → Lay flat* — it doesn't know which side is the patient. Rotate the model manually so the concave patient-facing surface sits against the plate. QuikBolus exports include the patient-facing-surface orientation hint — use it.



Step 2: Support settings (TPU bolus)

SETTING	VALUE	WHY
Support type	Tree (organic)	Less material, easier to remove than "normal". Marks TPU less.
Support style	Snug	Tighter contour, less waste.
Top Z gap	0.25 mm	TPU is the stickiest filament there is. Tighter than 0.25 = unremovable. 0.20 is too aggressive.

SETTING	VALUE	WHY
Interface layers	2	Smooth contact between support and part.
Interface pattern	Rectilinear interlaced	Comes off in clean strips.
Threshold angle	30°	Overhangs steeper than 30° from vertical get supported.
Support filament	Same TPU	Single-material print. (Dual-material PLA-on-TPU would release cleaner, but AMS+TPU isn't recommended — not worth it.)

Step 3: When in doubt, paint supports instead of auto

For an anatomical bolus, auto-generated supports often add material where you don't actually need it. Use Bambu Studio's **Paint Supports** tool to mark only the regions that overhang. The patient-contact surface (already on the plate, facing down) needs zero support. The outer dome typically needs minimal or no support — domes self-support up to ~45°.

Rule of thumb. A 5–10 mm thick bolus, oriented anatomy-side-down, usually needs supports **only at the perimeter rim** where it tapers to zero thickness, and possibly under deep concavities. The bulk of the part is self-supporting. If the auto-generator wants to fill the inside of your bolus with tree trunks, something's wrong — check orientation.

4. Bed prep — do these every time

- **Textured PEI plate** — never smooth PEI for TPU (sticks too well, you'll damage the contact surface removing it)
- Plate wiped with **isopropyl alcohol**; no glue stick, no hairspray, no Magigoo for TPU
- Filament **dried** if it's been open more than 2 weeks. Wet TPU = stringing + popping + voids. 60 °C for 6 hours in a filament dryer or food dehydrator
- AMS bypassed; filament loaded into the **rear PTFE port** directly from external spool holder
- **Flow calibration done** for this spool (Bambu Studio → Calibrate → Flow Dynamics + Flow Rate, ~15 min, once per spool)
- Bed leveled / auto-leveled within the last few prints; nozzle clean (no TPU bobble on the tip)

5. After the print — handling the finished bolus

1. **Remove from plate.** Flex the textured PEI plate gently — the bolus pops off. *Don't pry with a metal scraper*; you'll gouge the contact surface.
2. **Strip supports by hand.** Tree trunks pull off; interface layer peels. Stuck spots → flush cutters, never a knife.
3. **Inspect the patient-contact surface** under angled light. Slight plate texture transfer is fine and even desirable (small roughness helps skin adhesion). Hairline cracks or layer separation → reprint.
4. **Wash:** warm water + mild soap, rinse, isopropyl-wipe, air dry.
5. **Document** material lot, print profile name, print date, operator in the patient's chart. QuikBolus's audit record links to this.

Do not autoclave TPU 95A. It deforms above ~80 °C and the bolus loses its anatomical fit. Surface disinfection only (isopropyl, quaternary ammonium wipes). Single-patient use; discard after course completion or per your institutional policy.

6. When things go wrong

SYMPTOM	LIKELY CAUSE	FIX
Print starts then jams a few millimeters in	TPU bunching at extruder gear	Lower retraction to 0.3 mm; reduce pressure advance; verify AMS bypass
Stringy gaps between perimeters	Too fast or too hot	Outer wall → 20 mm/s; nozzle -5 °C
Bolus warps off plate at hour 2	First-layer adhesion / bed too cold	Brim 5 → 8 mm; bed +5 °C
Support fused permanently to part	Z gap too small	0.20 → 0.30 mm; consider lighter interface
Patient-contact surface looks rough, layer-lined	Bridging over support gaps	Re-orient: that surface goes ON the plate, not facing up
Print time "looks insane" (12+ hours)	TPU is just slow at quality settings	This is normal. Don't chase speed; sacrifice surface quality. Plan around the duration.
Surface popping / micro-voids	Wet filament	Dry the spool 6 h @ 60 °C, reprint
Stringing between distant features	Retraction set higher than TPU likes	0.5 mm and slow (30 mm/s); some stringing on TPU is unavoidable

7. Realistic print timing

TPU is slow. Plan accordingly — chasing speed costs you surface quality and adhesion, and there is no fast TPU.

BOLUS GEOMETRY	APPROX PRINT TIME	FILAMENT USED
5×7 cm × 5 mm thick (small head & neck)	4–6 hours	15–25 g
10×15 cm × 7 mm (breast / chest wall patch)	10–14 hours	60–90 g
20×30 cm × 7 mm (full chest wall)	18–26 hours	180–260 g
20×30 cm × 10 mm (full chest wall, thick)	24–34 hours	240–340 g

Schedule strategy: start chest-wall prints Friday evening; treat Monday. Head & neck patches start morning before clinic close; treat next morning. Always print one spare for each course in case of damage or fit issue.

8. Alternate workflow — print a mold, cast Flexibolus

For thicker bolus, build-up where water-equivalence matters most, or anatomy that wants the very softest conformance, an alternate workflow is to **3D-print a rigid mold** (PETG or PLA) and **cast the bolus from pourable Flexibolus** or similar tissue-equivalent silicone.

Trade-offs at a glance:

	DIRECT-PRINT TPU 95A	MOLD + FLEXIBOLUS CAST
Tissue-equivalent dose	Close (~1.21 g/cc)	Very close — Flexibolus is water-equivalent by design
Skin conformance	Good	Excellent
Production time	Print only (single material change)	Print mold + cure silicone (hours added) + demold
Reuse	Single-patient, single-course	Mold lasts many casts; bolus is single-patient
Operator time	Low	Higher — pour, degas, cure, demold
Surface finish	Plate-side textured	Glass-smooth (the mold's inner surface transfers)

Use TPU 95A direct print for most cases. Reach for the mold-and-cast workflow when you need thick build-up, when water-equivalence matters more than usual (e.g., electron skin treatments, superficial photon fields), or when the anatomy is unusually irregular and you need a softer-than-TPU result.

9. Further reading & references

- **Bambu Studio 101 — Beginners Guide to Bambu Slicer Software** (YouTube): if Bambu Studio's interface is brand-new to you, watch this first. youtube.com/watch?v=rZhy0J0mXBE
- **Bambu Lab TPU User Guide**: official guidance on AMS bypass, PTFE routing, and material handling. wiki.bambulab.com
- **AAPM TG-176** (Olch et al.): dosimetric considerations for surface bolus, including discussion of bolus material density.
- **QuikBolus Users Guide**: end-to-end workflow from DICOM RT Structure Set to STL, including audit/traceability fields. app.radiant-mpc.com/users-guides/quikbolus.pdf